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Module 4: Applied Visualizations

Project

- **Video:** Becoming an Independent Data Scientist
  1 min
- **Lab:** Project Description
  1h
- **Peer-graded Assignment:** Becoming an Independent Data Scientist
  2h
- **Review Your Peers:** Becoming an Independent Data Scientist
- **Reading:** Post-course Survey
  10 min
- **Reading:** 5 reasons to keep going
  3 min

Peer-graded Assignment: Becoming an Independent Data Scientist

You passed!
Congratulations! You earned 15 / 15 points. Review the feedback below and continue the course when you are ready.

Instructions **My submission** **Discussions**

Some of your fellow learners have submitted their reviews anonymously. All names are still visible to course instructors.

Analyzing the Passenger Pickup / Dropoff patterns for the NYC Yellow Taxi Dataset for April 2023

Submitted on September 21, 2024 Shareable Link

OVERALL ASSIGNMENT RUBRIC

In this assignment the focus is more on the mechanics of creating charts, so there is no requirement to create an enlightening chart as described by Cairo. But to explore the issue, please provide comments to the learner on the following:

1. Did the chart created teach you something new about the city/region/country? What did you learn?
2. Do you think this chart answers an interesting question? Why?
3. Name one thing which works and one which could be improved, and how would you improve it.

Muhammad Abdul Rosmi Alwi
Did the chart created teach you something new about the city/region/country? What did you learn? The chart provided new insights about the city/region/country, particularly in terms of how specific factors (e.g., population growth, economic trends, etc.) vary across different areas. I learned more about how these trends might impact development, social policies, or infrastructure planning in the region. Do you think this chart answers an interesting question? Why? Yes, the chart addresses an interesting question because it visualizes important data trends that could affect decisions or future planning. By presenting the information in an easy-to-digest format, the chart makes it easier to see patterns and relationships that might otherwise be overlooked. Name one thing which works and one which could be improved, and how would you improve it. One thing that works well is the clarity of the data representation, which makes the chart easy to understand at a glance. However, one area for improvement could be the inclusion of more detailed labels or interactive elements, such as tooltips, to provide additional context when hovering over specific data poires. This would make the chart more informative and engaging.

Anchita Jain
Yes. I hail from India and after this, I got insights about the taxi services in the USA. Yes I think the project was pretty good overall!

good

Translate to English

PROMPT

From time to time, Dr. Chris Brooks may want to share your work anonymously (via social media channels such as Twitter), to highlight some of the accomplishments of learners in this course.

If you agree that you are comfortable with instructor sharing your Assignment #4 as an example of great work, please type YES in the text box.

YES

PROMPT

State the region and the domain category that your data sets are about (e.g., Chaohu, China and sports or athletics).

The NYC Yellow Taxi Dataset is a public dataset that provides detailed records of taxi trips in New York City. Here is a breakdown of the region and domain category:

Region:

The dataset covers taxi trips specifically within New York City (NYC), including all five boroughs—Manhattan, Brooklyn, Queens, The Bronx, and Staten Island. Some trips may extend slightly outside of NYC for destinations such as nearby airports or surrounding areas.

Domain Category:

The domain of the dataset falls under transportation, specifically in the area of urban mobility and transportation analytics. It is often used for studying traffic patterns, transportation demand, ride-hailing services, public policy, and other areas of interest like urban planning, economics, and sustainability.

The data includes information such as pickup and dropoff locations, times, fare amounts, and passenger counts, among other features.

RUBRIC

Has the learner identified the region and the domain category that their data sets are about (e.g., Chaohu, China and sports or athletics)?

- 0 points
  The learner did not identify the region and domain category that their data sets are about.
- **2 points**
  The learner identified the region and domain category that their data sets are about.

MA AJ

PROMPT

Create a research question about the domain category and region that you identified.

How have the number of passengers (passenger density) in the yellow taxis for NYC for different TLC Taxi Zones (e.g., JFK Airport, Midtown Center, Penn Station/Madison Sq West etc.) in the month April of the year 2023 differ over with respect to different

1. zones
2. times (hours) in a day.
3. pickup vs. dropoff weekdays vs. weekends?

RUBRIC

Has the learner stated a research question about the domain category and region that they identified?

- 0 points
  The learner did not state a research question that related to the domain category and region they identified.
- **2 points**
  The learner did state a research question that related to the domain category and region they identified.

MA AJ

PROMPT

Provide at least two links to publicly accessible datasets. These could be links to files such as CSV or Excel files, or links to websites which might have data in tabular form, such as Wikipedia pages.

Here are couple of publicly accessible sources where we can find the **NYC Yellow Taxi Dataset** (for the year **2023** and month **April**):

1. **NYC OpenData**

   https://data.cityofnewyork.us/Transportation/2023-Yellow-Taxi-Trip-Data/4hki-vxec/data_preview

2. **TLC Trip Record Data**

   - https://www.nyc.gov/site/tlc/about/data-and-research.page
   - https://www.nyc.gov/assets/tlc/downloads/pdf/data_dictionary_trip_records_yellow.pdf

RUBRIC

Has the learner provided at least two links to available datasets? These could be links to files such as CSV or Excel files, or links to websites which might have data in tabular form, such as Wikipedia pages.

- 0 points
  The learner did not provide at least two working links to publicly available data sets (including links to CSV or Excel files).
- **2 points**
  The learner did provide links to at least two working links to publicly available data sets (including links to CSV or Excel files).

MA AJ

If you want to look at the learner's code, please do so here.

Muhammad Abdul Rosmi Alwi
The dataset links provided are accurate and easy to access. The data is well-structured, making it suitable for efficient analysis and visualization. Both datasets align well with the research question by providing comprehensive information, allowing for direct comparisons based on relevant variables such as location or category. This structure helps facilitate deeper insights and enables the creation of clear, informative visual comparisons.

Anchita Jain
NA

Translate to English

good

Translate to English

PROMPT

Upload an image which addresses your research question.

In addition to addressing your research question, this visual should address Cairo's principles of truthfulness, functionality, beauty, and insightfulness.

Analyzing the Passenger Pickup / Dropoff patterns for the NYC Yellow Taxi Dataset for April 2023
Analyzing the Passenger Pickup / Dropoff patterns for the NYC Yellow Taxi Dataset for April 2023
I have uploaded a pdf instead of an image, since I have multiple images needed to answer the research question. Please go through the uploaded pdf to find the images.

RUBRIC

Has the learner uploaded an image that addresses their stated question?

- 0 points
  The learner did not upload an image that addresses their stated question.
- **2 points**
  The learner uploaded an image that addresses their stated question.

MA AJ

PROMPT

Provide a short (1-2 paragraphs) justification of how your visual addresses your research question.

Total Pickup vs. Dropoff Count at different Zones

As can be seen from the geospatial visualizations (see the first pair of images in the pdf), the zones with maximum total passenger count (marked with darker color) for pickup (we get similar zones for dropoff too) are

- JFK Airport (LocationID 132)
- Midtown Center (LocationID 161)
- Upper East Side South (LocationID 237)

Total Pickup Count at different Zones in the Weekdays vs. Weekends

As can be seen from the second pair of geospatial images in the pdf uploaded earlier, the density of the passengers for pickup follows similar patterns at different zones for weekdays and weekends, with the main difference being that the total number of passengers picked up at different zones in the weekdays are much higher in the weekdays than in the weekends, as expected.

Total Pickup Passenger Count at different hours of a day in the Weekdays vs. Weekends

As can be seen from the lineplot image from the pdf uploaded, the peak hour for the total pickup count of the passengers is in between PM in the evening, whereas the total count drops significantly in the weekends, the trend being more or less the similar.

RUBRIC

Has the learner written a short (1-2 paragraph) justification of how their visualization addresses their question?

- 0 points
  The learner has not written a short (1-2 paragraph) justification of how their visualization addresses their question.
- **3 points**
  The learner has written a short (1-2 paragraph) justification of how their visualization addresses their question.

MA AJ

PROMPT

As this assignment is for the whole course, you must incorporate and defend the principles discussed in the first week, specifically, Cairo's principles of truth, beauty, function, and insight.

For each of the following prompts, please provide a response that links each principle to one or more elements of your visual.

- Describe your design choices for your visual in regards to Cairo's principle of **truthfulness**.
- Describe your design choices for your visual in regards to Cairo's principle of **beauty**.
- Describe your design choices for your visual in regards to Cairo's principle of **functionality**.
- Describe your design choices for your visual in regards to Cairo's principle of **insightfulness**.

Truthfulness

Principle: A visualization must be based on accurate, relevant, and factual data. It should not mislead the audience or distort the information.

- The entire data for the month April 2023 was used to answer to the research questions, there was no data hiding / cherry picking, all the data were use to compute aggregates.

Functionality

Principle: The visualization should be clear and easy to understand. It should effectively communicate the intended message without unnecessary complexity.

- For the geospatial visualizations, only the color coding property was used to depict the passenger density, it was simple and clear.

- For the lineplot, depicting the trend of the passenger counts over time used a simple design with clear labels, a legend, and distinct colors for weekdays and weekends. This helps viewers quickly grasp the trends without being overwhelmed by too much information.

Beauty

Principle: While aesthetics are important, they should not compromise the clarity and accuracy of the visualization. A beautiful design can engage the audience and make the information more appealing.

- The geospatial visualization to depict passenger densities at different zones used visually appealing elements like color coding using color maps, geospatial map overlay, and illustrations to make the data more engaging. These elements enhanced understanding of the zones.

- Using two different colors for showing trends for passenger counts over time for weekends vs weekdays should make the user ready grasp the message.

Insightfulness

Principle: A good visualization should provide insights that are not immediately obvious from the raw data. It should help the audience discover patterns, trends, or anomalies.

- From the geospatial colormaps, it was made clear that some zones (like JFK Airport) have much higher passenger density (both for pick and dorp) than the other zones.

- The trend shown as lineplot showed a clear difference between the weekdays vs. weekends trend in counts of passengers.

RUBRIC

Did the learner describe their design choices in regards to Cairo's principle of **truthfulness** (e.g., avoiding misleading representations) and relate them to one or more elements of their visual?

- 0 points
  No, the learner did not describe their design choices in regards to Cairo's principle of truthfulness (e.g., avoiding misleading representations) and did not relate them to one or more elements of their visual?
- **1 point**
  Yes, the learner did describe their design choices in regards to Cairo's principle of truthfulness (e.g., avoiding misleading representations) and relate them to one or more elements of their visual?

MA AJ

Did the learner describe their design choices in regards to Cairo's principle of **beauty** (e.g., using fonts, colors, and layout that are appropriate and aesthetically pleasing) and relate them to one or more elements of their visual?

- 0 points
  No, the learner did not describe their design choices in regards to Cairo's principle of beauty (e.g., using fonts, colors, and layout that are appropriate and aesthetically pleasing) and did not relate them to one or more elements of their visual?
- **1 point**
  Yes, the learner did describe their design choices in regards to Cairo's principle of beauty (e.g., using fonts, colors, and layout that are appropriate and aesthetically pleasing) and did relate them to one or more elements of their visual?

MA AJ

Did the learner describe their design choices in regards to Cairo's principle of **functionality** (e.g., choosing a charting type that is appropriate for the stated research question) and relate them to one or more elements of their visual?

- 0 points
  No, the learner did not describe their design choices in regards to Cairo's principle of functionality (e.g., choosing a charting type that is appropriate for the stated research question) and did not relate them to one or more elements of their visual?
- **1 point**
  Yes, the learner did describe their design choices in regards to Cairo's principle of functionality (e.g., choosing a charting type that is appropriate for the stated research question) and did relate them to one or more elements of their visual

MA AJ

Did the learner link Cairo's principle of **insightfulness** (i.e., producing a "eureka" or "aha" response in the viewer) with one or more elements of their visual?

- 0 points
  No, the learner did not make any connections between Cairo's principle of insightfulness with one or more elements of their visual.
- **1 point**
  Yes, the learner did make a connection between Cairo's principle of insightfulness with one or more elements of their visual.

MA AJ

PROMPT

Please upload your source code.

Analyzing the Passenger Pickup / Dropoff patterns for the NYC Yellow Taxi Dataset for April 2023
Analyzing the Passenger Pickup / Dropoff patterns for the NYC Yellow Taxi Dataset for April 2023
Analyzing the Passenger Pickup / Dropoff patterns for the NYC Yellow Taxi Dataset for April 2023

RUBRIC

If you want to look at the learner's code, we recommend that you open it through the Jupyter notebook system on the Coursera platform.

Muhammad Abdul Rosmi Alwi
To review the learner's code, it is recommended to open the Jupyter notebook system directly on the Coursera platform. This allows for a seamless experience in analyzing and executing the code in an interactive environment. By using Jupyter on Coursera, you can ensure that the code runs within the same environment it was created in, minimizing potential compatibility issues. Additionally, Jupyter provides powerful tools for visualizing data, running individual code cells, and making iterative improvements, making it the ideal platform for reviewing and experimenting with the learner's work.

Anchita Jain
NA

Translate to English

good

Translate to English

Start new attempt

Comments
Comments left for the learner are visible only to that learner and the person who left the comment.

SD Share your thoughts...

Like **Dislike** **Report an issue**